Appendix: How smoothing the affinity matrix affects neighborhood preservation in t-SNE

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This appendix provides dataset preprocessing details and full experimental results for all three datasets: MNIST, mouse cortex, and UCI Adult. The main paper reports results on MNIST only due to space constraints. Section A presents preprocessing details. Section B presents detailed effective perplexity analysis for all three datasets. Sections C–G present the full results corresponding to each experiment in the main paper, in the same order: effect of smoothing on local neighborhood preservation, smoothing versus increasing perplexity, sensitivity to γ and ρ, effect of smoothing on global structure preservation, and comparison with alternative affinity constructions.

A Dataset preprocessing details

MNIST. MNIST [3] contains images of handwritten digits (n = 70,000, m = 784). Pixel values are scaled to $[0, 1]$ and the data are reduced to 50 principal components before constructing t-SNE affinities, following the preprocessing recommended by Kobak and Berens [2].

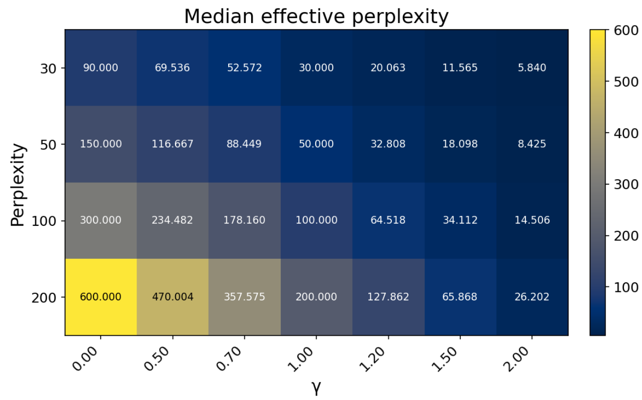
UCI Adult. UCI Adult [1] is a tabular dataset of US census data (n = 48,842, $m = 14$). Numerical variables are standardized and categorical variables are one-hot encoded before constructing t-SNE affinities.

Mouse cortex. The mouse cortex dataset [4] contains single-cell RNA sequencing data from adult mouse cortex, grouped into 133 clusters with a strong hierarchical organization (n = 23,822). We use the same preprocessed data as Kobak and Berens [2], which includes sequencing-depth normalization, feature selection, log-transformation, and dimensionality reduction to 50 principal components.

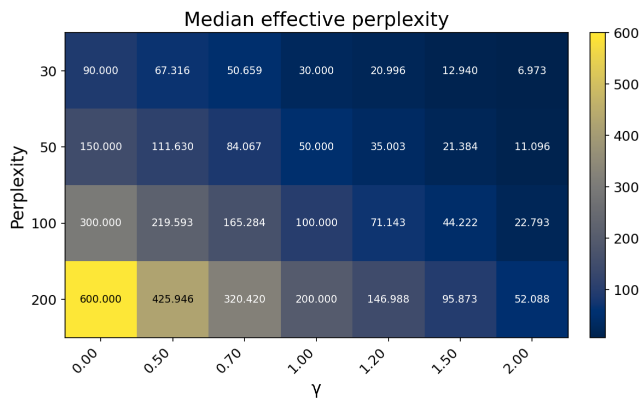
B Effective perplexity analysis

Figure 1 shows the median effective perplexity on MNIST, mouse cortex, and Adult across a wider range of γ and ρ settings. When $\gamma = 1.0$, the effective perplexity matches ρ exactly. Sharpening reduces effective perplexity by concentrating the distribution and lowering its entropy, while smoothing increases it. The extreme case $\gamma = 0$ produces a uniform distribution, giving an effective perplexity of $3 \cdot \rho$, while $\gamma = 2.0$ at $\rho = 30$ reduces it to only 5.840 on MNIST.

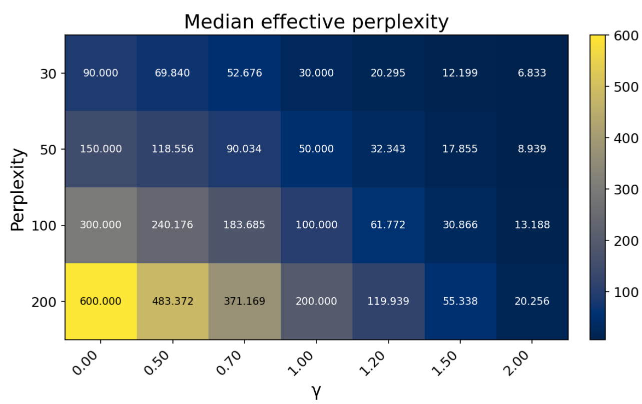
Figure 2 shows the per-point change in effective perplexity after smoothing ($\gamma = 0.7$, $\rho = 30$) across all three datasets. For the top-5 conditional mass (left column), the pattern is consistent and clear across all datasets: points with sharper affinity rows receive a larger increase in effective perplexity, with Spearman correlations of 0.667, 0.797, and 0.789 for MNIST, mouse cortex, and Adult respectively. For the bandwidth σ_i (right column), the relationship is less consistent. For MNIST and Adult, points in sparser regions tend to receive a larger increase in effective perplexity, with Spearman correlations of 0.488 and 0.678. For mouse cortex, however, the relationship is essentially absent (Spearman $\rho = -0.030$), which may reflect the more complex density structure of that dataset.



(a) MNIST

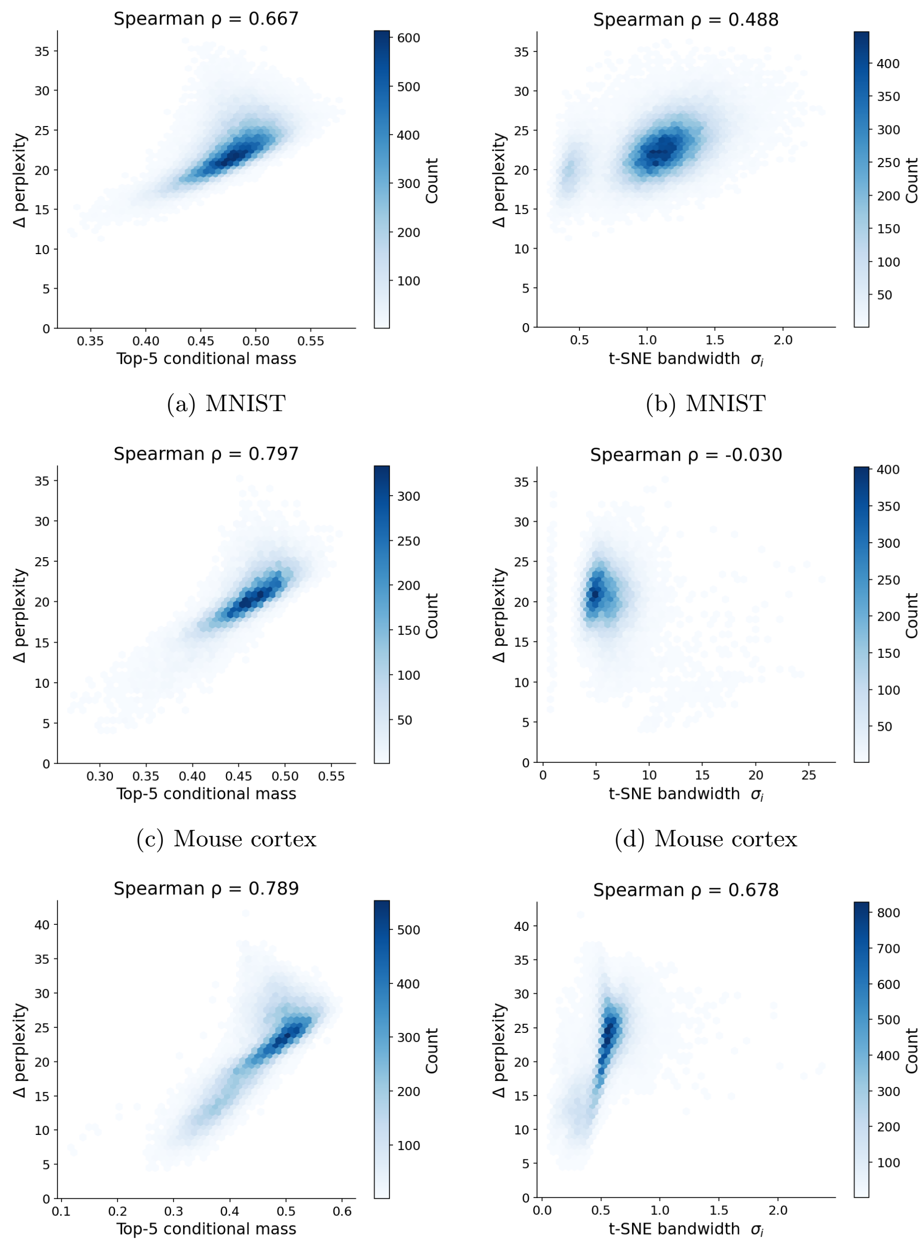


(b) Mouse cortex



(c) Adult

Fig. 1: Median effective perplexity across ρ and γ settings on MNIST (a), mouse cortex (b), and Adult (c).



(a) MNIST (b) MNIST

(c) Mouse cortex (d) Mouse cortex

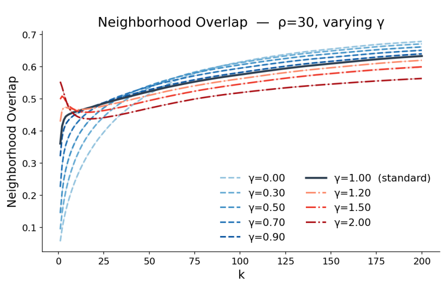
(e) Adult (f) Adult

Fig. 2: Per-point change in effective perplexity after smoothing ($\gamma = 0.7$, $\rho = 30$). (left) Versus top-5 conditional mass $M_i(5)$. (right) Versus t-SNE bandwidth σ_i across three datasets.

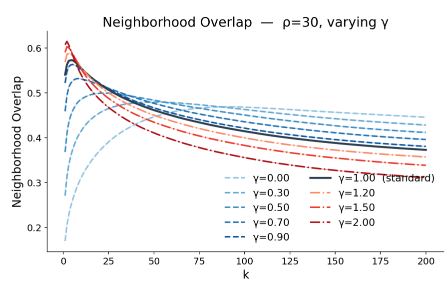
C Effect of smoothing on local neighborhood preservation

Figure 3 shows $NO@k$ for $k = 1, \ldots, 200$ at perplexity $\rho = 30$ for mouse cortex and Adult. Smoothed variants ($\gamma < 1$) score better at higher values of k, with the gap growing as k increases, while sharpened variants ($\gamma > 1$) score better at small k. For small values of k, sharpened variants ($\gamma > 1$) outperform standard t-SNE. This is because sharpening concentrates more probability mass on the nearest neighbors, giving them stronger attractive force during optimization. For $k > 10$, this behavior reverses: smoothed variants ($\gamma < 1$) consistently outperform both standard t-SNE and sharpened variants, with the gap growing as k increases. For example in mouse cortex data, $\gamma = 0.5$ reaches approximately 0.6 at $k = 100$ compared to 0.57 for standard t-SNE and 0.51 for $\gamma = 2.0$. We observed the same crossover pattern on all three datasets, confirming that this is not specific to MNIST. This happens because smoothing redistributes probability mass toward mid-ranked neighbors, giving them more influence during optimization rather than letting the nearest neighbors dominate.

Figure 4 shows the embeddings for mouse cortex and Adult; the MNIST embedding is shown in the main paper, where the effect of smoothing is most visible, with clusters more separated and digit classes that overlap in standard t-SNE pulled apart. For mouse cortex, the overall hierarchical structure is preserved in both embeddings, with the smooth version showing slightly more separated clusters. For Adult, both embeddings look similar, which is expected: the dataset does not have strong cluster structure, so smoothing has less visible effect on the embedding layout, even though the quantitative neighborhood preservation metrics still show improvement.

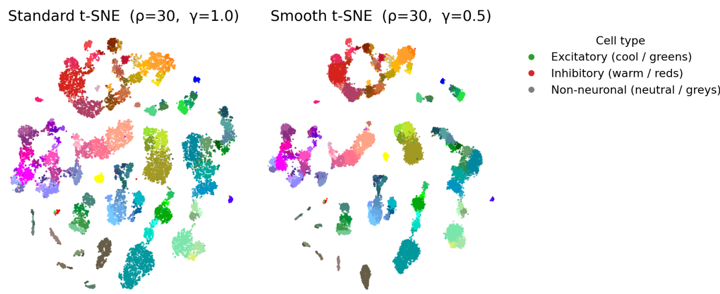


(a) mouse cortex

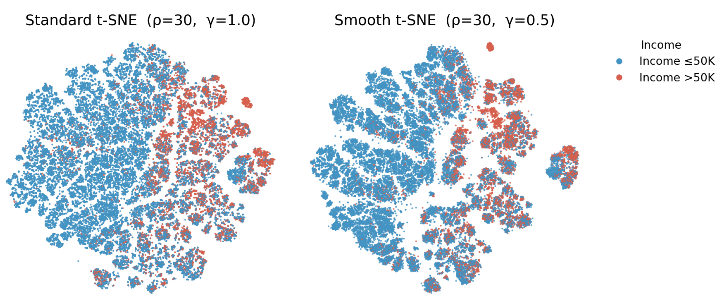


(b) adult

Fig. 3: $NO@k$ for $k = 1, \ldots, 200$ at $\rho = 30$ with varying γ, on mouse cortex (a) and Adult (b). The MNIST result is shown in the main paper.



(a) mouse cortex



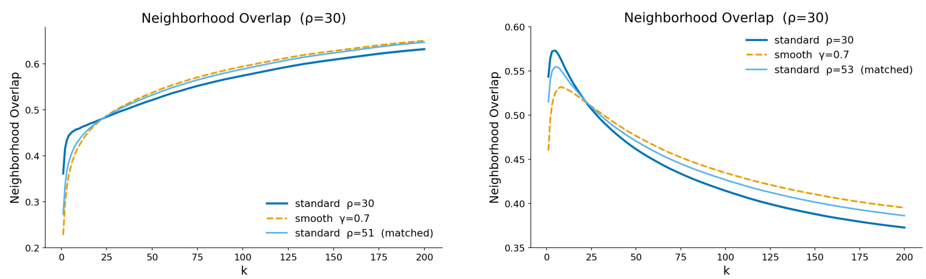
(b) adult

Fig. 4: Standard t-SNE versus smoothed t-SNE ($\gamma = 0.5$) on mouse cortex (a) and Adult (b). The MNIST embedding is shown in the main paper.

D Smoothing versus increasing perplexity

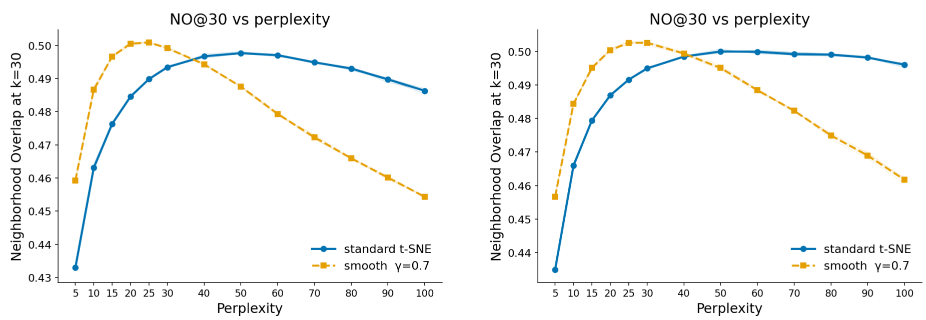
Figure 5 shows the mouse cortex and Adult results; together with the MNIST result in the main paper, this confirms across all three datasets that smoothing is not equivalent to increasing the global perplexity. In all cases, smoothed t-SNE ($\gamma = 0.7$) outperforms matched-perplexity standard t-SNE at broader k, despite having the same mean effective perplexity. This confirms that the point-dependent effective perplexities produced by smoothing lead to different and better mid-local preservation than simply increasing the global perplexity.

Figure 6 shows $NO@30$ while varying perplexity for standard and smoothed t-SNE (with constant $\gamma = 0.7$, not tuned) on mouse cortex and Adult (the MNIST result is shown in the main paper). The same pattern holds in all cases. Smoothing strongly affects which perplexity is optimal: for standard t-SNE the best perplexity is around 40–50 depending on the dataset, while smoothed t-SNE peaks at a notably lower perplexity of around 15–25. In all three datasets, the highest $NO@30$ reached by smoothed t-SNE across perplexities exceeds the highest reached by standard t-SNE. However, at higher perplexities smoothed t-SNE degrades more sharply, consistent with the finding that smoothing has adverse effect when the affinity rows are already relatively smooth.



(a) Mouse cortex (b) Adult

Fig. 5: $NO@k$ comparison between standard t-SNE, smoothed t-SNE ($\gamma = 0.7$), and matched-perplexity standard t-SNE on mouse cortex (a) and Adult (b). The MNIST result is shown in the main paper.



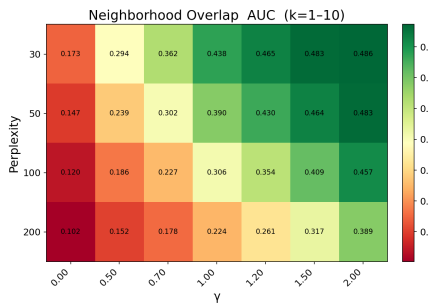
(a) Mouse cortex (b) Adult

Fig. 6: $NO@30$ as a function of perplexity for standard t-SNE and smoothed t-SNE ($\gamma = 0.7$, not tuned) on mouse cortex (a) and Adult (b). Smoothed t-SNE peaks at a lower perplexity than standard t-SNE and attains a higher peak $NO@30$. The MNIST result is shown in the main paper.

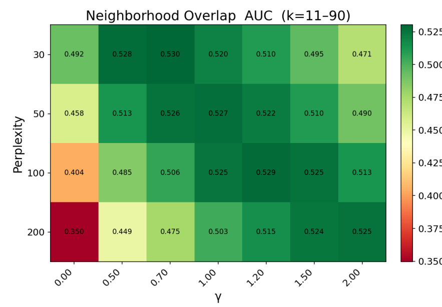
E Sensitivity to γ and ρ

Figure 7 shows the sensitivity results for mouse cortex and Adult; together with the MNIST results in the main paper, they cover all three datasets. For near-local preservation (left column), the pattern is consistent: sharpening improves performance and smoothing hurts it, regardless of dataset or perplexity. This confirms the finding in the main paper holds generally.

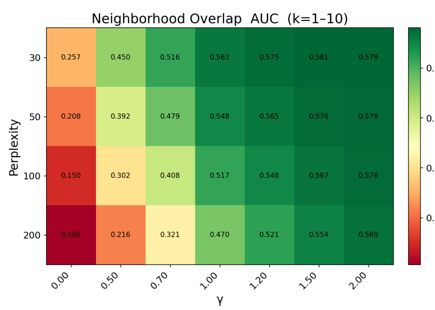
For mid-local preservation (right column), the pattern is mostly consistent for MNIST and mouse cortex, where moderate smoothing helps at lower perplexities. Adult shows the same shift, but weaker. Smoothing ($\gamma = 0.7$) is best only at low perplexity ($\rho = 30$ and 50). At $\rho = 100$ and $\rho = 200$ the best result moves to $\gamma = 1.0$–1.2, and $\gamma = 0.7$ drops below standard t-SNE.



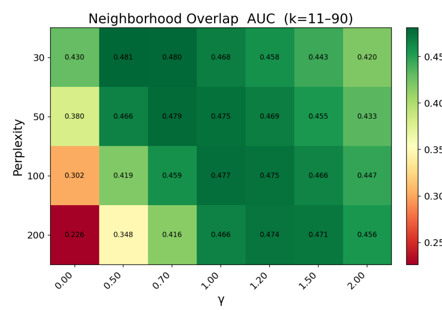
(a) Near-local AUC, Mouse cortex



(b) Mid-local AUC, Mouse cortex



(c) Near-local AUC, Adult

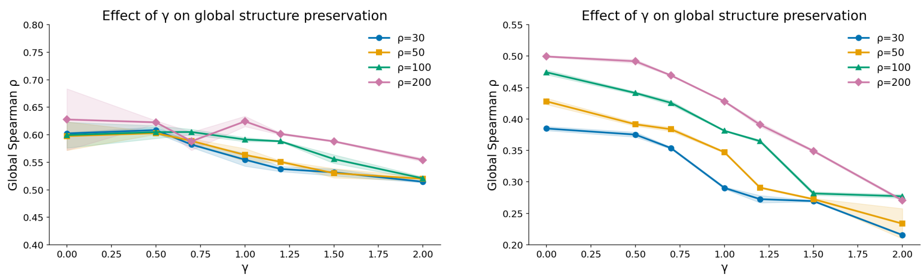


(d) Mid-local AUC, Adult

Fig. 7: Sensitivity of neighborhood preservation to γ and ρ on mouse cortex (a, b) and Adult (c, d). The MNIST result is shown in the main paper.

F Effect of smoothing on global structure preservation

Figure 8 shows the effect on global structure preservation for mouse cortex and Adult. The main finding holds across all three datasets (including MNIST in the main paper): sharpening consistently degrades global structure while smoothing generally improves it, most clearly at higher perplexities. Adult shows the strongest absolute values and clearest effect, while MNIST shows the weakest effect, consistent with MNIST having strong local cluster structure that dominates the optimization regardless of γ.

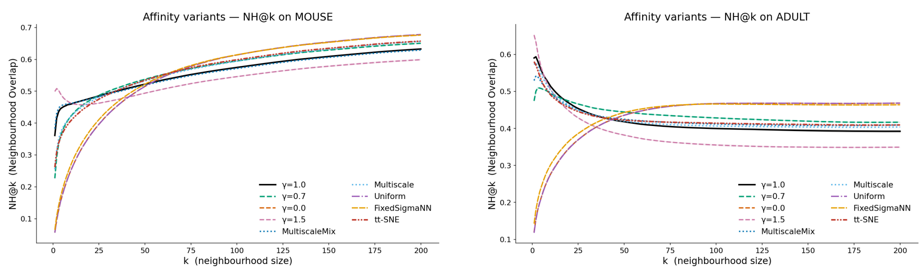


(a) Mouse cortex (b) Adult

Fig. 8: Effect of γ on global structure preservation on mouse cortex (a) and Adult (b). The MNIST result is shown in the main paper.

G Comparison with alternative affinity constructions

Figure 9 shows the affinity variant comparison on mouse cortex and Adult. The overall pattern from MNIST holds across both datasets: $\gamma = 1.5$ leads at small k and $\gamma = 0.7$ outperforms standard t-SNE in the mid-local range. On mouse cortex, the multiscale methods are more competitive than on MNIST, tracking closely with $\gamma = 0.7$ in the mid-to-large range, while $\gamma = 0.0$, `Uniform`, and `FixedSigmaNN` again dominate at large k. Adult shows a distinct pattern: $\gamma = 0.7$ maintains its advantage across a wider range of k, remaining competitive with $\gamma = 0.0$ and `FixedSigmaNN` even at large k, while the multiscale methods offer little improvement over standard t-SNE. Across all datasets, $\gamma = 1.5$ ends with the lowest $NO@k$ at large k, with the drop most pronounced on Adult.



(a) Mouse cortex (b) Adult

Fig. 9: $NO@k$ for $k = 1, \ldots, 200$, comparing γ variants against alternative affinity constructions on mouse cortex (a) and Adult (b). The overall pattern from MNIST holds, with $\gamma = 0.7$ outperforming other methods in the mid-local range. On Adult, $\gamma = 0.7$ maintains its advantage more broadly, while on mouse cortex the multiscale methods are more competitive at larger k.

References

1. Becker, B., Kohavi, R.: Adult data. UCI Machine Learning Repository (1996)
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4. Tasic, B., et al.: Shared and distinct transcriptomic cell types across neocortical areas. Nature **563**, 72–78 (2018)